Material requirements

Parts	**DB-FOOT-SOLE-L, DB-FOOT-SOLE-R (soles, elastomer); optional DB-FOOT-L, DB-FOOT-R (rigid feet)**
Files	`sole_left.step`, `sole_right.step`, `foot_left.step`, `foot_right.step` (in `duckbuddy-feet-cad.zip`; STL copies on request), drawing `duckbuddy-feet-drawing.pdf`
Contact	Rafael Caricio, rafael@caricio.com

1. Use of the parts

The soles are the ground contact of a small two-legged walking robot. Each sole is a thin cup (53.6 × 40.9 × 13.3 mm, walls 1.2–2.1 mm, floor 2.0 mm) that is stretched over a rigid printed foot. The sole must grip the floor, damp the impact of each step and stay on the foot.

- Floors: wood, laminate, tile and short carpet, indoors, dry.
- Temperature 10–40 °C. No oil, fuel or solvents.
- Load: repeated impacts and short slides at every step, low load (small robot).

2. Soles: cast polyurethane elastomer

Property	Requirement
Process	Vacuum casting in a silicone mould (or the process Yellowhammer recommends for this hardness and wall thickness).
Hardness	**Shore 60A ± 5** (target 55–60A, never above 65A). Measured to ISO 48-4 / ASTM D2240 type A at 23 °C after full post-cure.
Tensile strength	≥ 2 MPa
Elongation at break	≥ 200 %
Tear strength	≥ 12 kN/m (ISO 34-1 / ASTM D624 die C)
Colour	**Bright orange**, for example RAL 2004 (pure orange). Opaque, coloured through (pigmented, not painted). The same shade as the feet if you also make them.
Surface	Matte, as cast. Clean: no release-agent film (it lowers grip, and the inside is glued to the foot).

Reference material. The robot was tuned with Polytek **PT Flex 60** in mind: Shore A 55–60, tensile 2.1–2.4 MPa, elongation 222–235 %, tear (die C) 70–71 pli (about 12.4 kN/m), linear shrinkage about 0.25 % (Polytek data sheets; the ranges cover the 2008 sheet and the 2021 technical bulletin). PT Flex 60 is an open-pour RTV rubber, so it is named only as the property target. A vacuum-casting PU elastomer set to about 60A is preferred, for example SikaBiresin PX840 at 60–65A, Hei-Cast 8400 at 60A or ALCHEMIX VC 3300 at 60A. Please name the resin you will use and send its datasheet with the quote.

Hardness is the key value. Please do not change the hardness or the material without written agreement (your general terms allow small hardness deviations unless agreed in writing; this document is that written requirement).

3. Feet (optional line item): rigid

Property	Requirement
Process	SLS printing, dyed. MJF only if you can make it bright orange (standard MJF PA12 is grey). Vacuum casting in a rigid PU (Shore D 80 or more), pigmented orange, is also fine if it is cheaper for the quantity.
Material	PA12 (for example HP 3D HR PA 12), or a rigid PU of similar strength.
Colour	**Bright orange**, the same shade as the soles (for example RAL 2004). Dyed or pigmented through, not painted.

Screws	The part takes M2 self-tapping screws in its modelled pilot holes, so the material must not crack when screwed.

4. Quantities

One set is one left and one right part.

- Soles: please quote **2, 5 and 10 sets**, with the master and mould cost on its own line.
- Feet (optional): **2 sets**, quoted separately from the soles.

5. Please send with the quote

- The resin name and datasheet for the soles, and the hardness you will cast (60A ± 5), confirmed in the quotation itself.
- Price per set for each quantity in the request, with the mould/master cost shown on its own line.
- Lead time, and how many castings one silicone mould gives for this part.